Mathematical Addendum: Response to Critique on the Feasible Region Ω and Stochastic Dynamics

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Context

We thank the reviewer for identifying three structural gaps in the original existence and invariance argument for the feasible region $\Omega \subset \mathbb{R}^{128}$ under stochastic updates $x(t+1) = f_0(x(t)) + \sigma\eta(t)$. All three points have been formally resolved and integrated into the revised manuscript (Sec. 3–4). Below is the consolidated mathematical statement.

1. Convexity of Ω

The global region $\Omega = \bigcap_i \{x : A_i(x)\}$ is not convex: the entropy bound $H(x) \leq H_{\max}$ is an upper level set of a concave function, and the ethical exclusion A_0 removes a half-space. A direct application of Brouwer's fixed-point theorem on Ω is therefore unjustified.

Resolution. We isolate the *contractive core*

$$\Omega_A = \{x \in \Omega : \cos(x, s) \geq \theta_A\}, \quad \theta_A = 0.88,$$

which is a spherical cap nested inside the viable region $\Omega \supseteq \{\cos \geq 0.70\}$. Ω_A is compact and convex. Existence of an invariant measure is now carried on Ω_A via Krasnosel'skiĭ's fixed-point theorem for sums of a contraction and a compact continuous perturbation (Krasnosel'skiĭ [1955]). The contraction hypothesis is discharged by measurement: $\lambda_{\max} = \ln(0.99) < 0$ (EXP-LY-01).

2. Continuity and Invariance of f_0

Forcing $f_0(\Omega) \subseteq \Omega$ by projection would distort the natural dynamics and obscure failure modes. We do not impose invariance globally.

Resolution. The deterministic map f_0 exhibits layered behavior:

- On Ω_A: f_0 is a contraction with Lipschitz constant $L < 1$ (measured $\lambda_{\max} < 0$).
- On $\Omega \setminus \Omega_A$: f_0 is non-expansive ($L \leq 1$) and excursions are bounded by a finite recovery time τ_{rec}.

No artificial clamping is applied. The geometry of the Lyapunov function $L(x)$ ensures that states perturbed outside Ω_A are naturally re-contracted into the core without leaving Ω (Remark 2, Sec. 3). Invariance is thus a *measured dynamical property*, not an imposed algebraic constraint.

3. Stochastic Perturbations $\sigma \cdot \eta(t)$

Even if $f_0 : \Omega \to \Omega$ were invariant, the additive noise $\sigma\eta(t)$ with unbounded support breaks pointwise invariance: $x(t+1) \notin \Omega$ with non-zero probability.

Resolution. We replace pointwise invariance with *stochastic viability* (Aubin [1991]) and the Lyapunov drift condition

$$\mathbb{E}[L(x(t+1)) - L(x(t)) \mid x(t) \in \Omega] \leq 0,$$

which is exactly the geometric-ergodicity condition for the induced Markov operator (Meyn & Tweedie [2009]). Confinement is empirical: EXP-NS-01 drives the kernel with sustained bounded noise at every tick and shows

$$P\big(\cos(x(t), s) \geq \theta_{\text{viable}}\big) = 1.0000 \quad \forall \sigma \in [0, 2.0\|x\|].$$

The viability margin $m(\sigma)$ degrades smoothly and saturates at ≈ 0.33; there is no finite noise ceiling. The system does not rely on hard boundaries; it relies on negative expected drift that recontracts excursions in finite time.

Integration with the Manuscript

The three objections are resolved in the current revision:

1. **Sec. 3, Def. 2 & Thm. 1:** Existence argument carried on the convex core Ω_A via Krasnosel'skiĭ; global non-convexity handled by confinement.
2. **Sec. 4, Thm. 2 & EXP-LY-01:** Convergence in expectation via measured Lyapunov drift ($\lambda_{\max} = -0.010050$), not assumed invariance.
3. **Sec. 6.4, EXP-NS-01:** Sustained-noise margin curve replaces the borrowed impulsive proxy; confinement $P = 1.0000$ up to $\sigma = 2.0\|x\|$ verifies stochastic viability empirically.

The existence proof is now *measured, not postulated*: contraction, confinement, and ergodicity are discharged by hardware experiments rather than asserted algebraically. Brouwer's theorem is no longer invoked; the negative Lyapunov exponent and the viability-theoretic reading of Ω carry the argument.

References

[1] J.-P. Aubin, *Viability Theory*, Birkhäuser, 1991.

[2] M. A. Krasnosel'skiĭ, "Two remarks on the method of successive approximations," *Uspekhi Mat. Nauk*, 10(1):123–127, 1955.

[3] S. P. Meyn and R. L. Tweedie, *Markov Chains and Stochastic Stability*, Cambridge Univ. Press, 2nd ed., 2009.